

Bar prints 100mm @ 100%

Santen

COMMERCIAL ARTWORK

Description:

Cosopt 20 mg/ml + 5 mg/ml (MY) 5 ml Leaflet

CMO Item Number:

N/A

CMO SPEC Number:

N/A

GLAMS Part Number:

CAW-5799-02

Technical Drawing No:

N/A

Dimension:

387 X 240 mm

Pharma Code:

7852

2D matrix:

N/A

Date:

18-05-2023

Market:

MY

Material Code:

754751

Artwork Version Number:

03

Schawk Job No.:

108718471/403302982

Printable Colours:

Santen PI Blue

Non Print:

Profile

Technical Info

SGK

SGK is a Matthews International Corporation

2582
NO TITLING HERE

75475103

LOCAL PRODUCT CIRCULAR
Sterile Ophthalmic Solution

COSOPT®

(dorzolamide hydrochloride and timolol maleate ophthalmic solution)

I. THERAPEUTIC CLASS

COSOPT® Ophthalmic Solution (dorzolamide hydrochloride and timolol maleate) is the first combination of a topical carbonic anhydrase inhibitor and a topical beta-adrenergic receptor blocking agent.

II. CLINICAL PHARMACOLOGY

Ila. Mechanism of Action

COSOPT is comprised of two components: dorzolamide hydrochloride and timolol maleate. Each of these two components decreases elevated intraocular pressure by reducing aqueous humor secretion, but does so by a different mechanism of action.

Dorzolamide hydrochloride is a potent inhibitor of human carbonic anhydrase II. Inhibition of carbonic anhydrase in the ciliary processes of the eye decreases aqueous humor secretion, presumably by slowing the formation of bicarbonate ions with subsequent reduction in sodium and fluid transport. Timolol maleate is a nonselective beta-adrenergic receptor blocking agent that does not have significant intrinsic sympathomimetic, direct myocardial depressant, or local anesthetic (membrane-stabilizing) activity. The combined effect of these two agents results in additional intraocular pressure (IOP) reduction compared to either component administered alone.

Following topical administration, COSOPT reduces elevated intraocular pressure, whether or not associated with glaucoma. Elevated intraocular pressure is a major risk factor in the pathogenesis of optic nerve damage and glaucomatous visual field loss. The higher the level of intraocular pressure, the greater the likelihood of glaucomatous visual field loss and optic nerve damage. COSOPT reduces intraocular pressure without the common side effects of miotics such as night blindness, accommodative spasm and pupillary constriction.

Ilb. Pharmacokinetics / Pharmacodynamics

Dorzolamide Hydrochloride

Unlike oral carbonic anhydrase inhibitors, topical administration of dorzolamide hydrochloride allows for the drug to exert its effects directly in the eye at substantially lower doses and therefore with less systemic exposure. In clinical trials, this resulted in a reduction in IOP without the acid-base disturbances or alterations in electrolytes characteristic of oral carbonic anhydrase inhibitors.

When topically applied, dorzolamide reaches the systemic circulation. To assess the potential for systemic carbonic anhydrase inhibition following topical administration, drug and metabolite concentrations in RBCs and plasma and carbonic anhydrase inhibition in RBCs were measured. Dorzolamide accumulates in RBCs during chronic dosing as a result of selective binding to CA-II while extremely low concentrations of free drug in plasma are maintained. The parent drug forms a single N-desethyl metabolite that inhibits CA-II less potently than the parent drug but also inhibits a less active isoenzyme (CA-I). The metabolite also accumulates in RBCs where it binds primarily to CA-I. Dorzolamide binds moderately to plasma proteins (approximately 33%). Dorzolamide is primarily excreted unchanged in the urine; the metabolite is also excreted in urine. After dosing ends, dorzolamide washes out of RBCs nonlinearly, resulting in a rapid decline of drug concentration initially, followed by a slower elimination phase with a half-life of about four months.

When dorzolamide was given orally to simulate the maximum systemic exposure after long term topical ocular administration, steady state was reached within 13 weeks. At steady state, there was virtually no free drug or metabolite in plasma; CA inhibition in RBCs was less than that anticipated to be necessary for a pharmacological effect on renal function or respiration. Similar pharmacokinetic results were observed after chronic, topical administration of dorzolamide hydrochloride. However, some elderly patients with renal impairment (estimated CrCl 30-60 milliliter/min) had higher metabolite concentrations in RBCs, but no meaningful differences in carbonic anhydrase inhibition and no clinically significant systemic side effects were directly attributable to this finding.

Timolol Maleate

In a study of plasma drug concentration in six subjects, the systemic exposure to timolol was determined following twice daily topical administration of timolol maleate ophthalmic solution 0.5%. The mean peak plasma concentration following morning dosing was 0.46 ng/mL and following afternoon dosing was 0.35 ng/mL.

III. INDICATIONS

COSOPT is indicated in the treatment of elevated intraocular pressure (IOP) in patients with ocular hypertension and open-angle glaucoma when concomitant therapy is appropriate.

IV. DOSAGE AND ADMINISTRATION

The dose is one drop of COSOPT in the affected eye(s) two times daily.

When substituting COSOPT for another ophthalmic antiglaucoma agent(s), discontinue the other agent(s) after proper dosing on one day, and start COSOPT on the next day.

If another topical ophthalmic agent is being used, COSOPT and the other agent should be administered at least ten minutes apart.

Safety and efficacy in pediatric patients below the age of 2 years have not been established. (For information regarding use in pediatric patients ≥2 years of age see section IX)

When using nasolacrimal occlusion or closing the eyelids for 2 minutes, the systemic absorption is reduced. This may result in an increase in local activity.

V. CONTRAINDICATIONS

COSOPT is contraindicated in patients with:

- reactive airway disease, bronchial asthma or a history of bronchial asthma, or severe chronic obstructive pulmonary disease
- sinus bradycardia, sino-atrial block, second or third degree atrioventricular block, overt cardiac failure, cardiogenic shock
- hypersensitivity to any component of this product

The above are based on the components and are not unique to the combination.

VI. PRECAUTIONS

As with other topically-applied ophthalmic agents, this drug may be absorbed systemically. The timolol component is a betablocker. Therefore, the same types of adverse reactions found with systemic administration of beta-blockers may occur with topical administration.

Cardio-respiratory Reactions

Because of the timolol maleate component, cardiac failure should be adequately controlled before beginning therapy with COSOPT. Patients with a history of cardiovascular disease, including cardiac failure, should be watched for signs of deterioration of these diseases, and pulse rates should be checked.

Due to its negative effect on conduction time, beta-blockers should be given with caution to patients with first degree heart block.

Respiratory reactions and cardiac reactions, including death due to bronchospasm in patients with asthma and rarely death in association with cardiac failure, have been reported following administration of timolol maleate ophthalmic solution.

In patients with mild/moderate chronic obstructive pulmonary disease (COPD), COSOPT should be used with caution, and only if the potential benefit outweighs the potential risk.

Vascular Disorders

Patients with severe peripheral circulatory disturbance/disorders (e.g. severe forms of Raynaud's disease or Raynaud's syndrome) should be treated with caution.

Masking of Hypoglycemic Symptoms in Patients with Diabetes Mellitus

Beta-adrenergic blocking agents should be administered with caution in patients subject to spontaneous hypoglycemia or to diabetic patients (especially those with labile diabetes) who are receiving insulin or oral hypoglycemic agents. Beta-adrenergic blocking agents may mask the signs and symptoms of acute hypoglycemia.

Masking of Thyrotoxicosis

Beta-adrenergic blocking agents may mask certain clinical signs of hyperthyroidism (e.g., tachycardia). Patients suspected of developing thyrotoxicosis should be managed carefully to avoid abrupt withdrawal of beta-adrenergic blocking agents which might precipitate a thyroid storm.

Surgical Anesthesia

The necessity or desirability of withdrawal of beta-adrenergic blocking agents prior to major surgery is controversial. If necessary during surgery, the effects of beta-adrenergic blocking agents may be reversed by sufficient doses of adrenergic agonists.

Renal and Hepatic Impairment

COSOPT has not been studied in patients with severe renal impairment (CrCl<30milliliter/min). Because dorzolamide hydrochloride and its metabolite are excreted predominantly by the kidney, COSOPT is not recommended in such patients.

COSOPT has not been studied in patients with hepatic impairment and therefore should be used with caution in such patients.

Immunology and Hypersensitivity

As with other topically-applied ophthalmic agents, this drug may be absorbed systemically. The dorzolamide component is a sulfonamide. Therefore, the same types of adverse reactions found with systemic administration of sulfonamides may occur with topical administration, such as Stevens-Johnson syndrome and toxic epidermal necrolysis. If signs of serious reactions or hypersensitivity occur, discontinue use of this preparation.

In clinical studies, local ocular adverse effects, primarily conjunctivitis and lid reactions, were reported with chronic administration of dorzolamide hydrochloride ophthalmic solution. Some of these reactions had the clinical appearance and course of an allergic-type reaction that resolved upon discontinuation of drug therapy. Similar reactions have been reported with COSOPT. If such reactions are observed, discontinuation of treatment with COSOPT should be considered.

While taking β-blockers, patients with a history of atopy or a history of severe anaphylactic reaction to a variety of allergens may be more reactive to accidental, diagnostic, or therapeutic repeated challenge with such allergens. Such patients may be unresponsive to the usual doses of epinephrine used to treat anaphylactic reactions.

Concomitant Therapy

There is a potential for an additive effect on the known systemic effects of carbonic anhydrase inhibition in patients receiving oral and topical carbonic anhydrase inhibitors concomitantly. The concomitant administration of COSOPT and oral carbonic anhydrase inhibitors has not been studied and is not recommended.

Patients who are already receiving a beta-adrenergic blocking agent systemically and who are given COSOPT should be observed for a potential additive effect either on the intraocular pressure or on the known systemic effects of beta-blockade. The use of two topical beta-adrenergic blocking agents is not recommended.

Other

The management of patients with acute angle-closure glaucoma requires therapeutic interventions in addition to ocular hypotensive agents. COSOPT has not been studied in patients with acute angle-closure glaucoma.

Choroidal detachment has been reported with administration of aqueous suppressant therapy (e.g., timolol, acetazolamide, dorzolamide) after filtration procedures.

There is an increased potential for developing corneal edema in patients with low endothelial cell counts. Precautions should be used when prescribing COSOPT to this group of patients.

Contact Lens Use

COSOPT contains the preservative benzalkonium chloride, which may be deposited in soft contact lenses; therefore, COSOPT should not be administered while wearing these lenses. The lenses should be removed before application of the drops and not be reinserted earlier than 15 minutes after use.

MUTAGENESIS, CARCINOGENESIS AND IMPAIRMENT OF FERTILITY

Dorzolamide Hydrochloride

In a 2 year study of dorzolamide hydrochloride administered orally to male and female rats, urinary bladder papillomas were seen in male rats in the highest dosage group of 20mg/kg/day. No treatment related tumors were seen in a 21 months study in male and female mice given oral doses up to 75 and 37.5mg/kg/day, respectively.

The increased incidence of urinary bladder papillomas seen in the high-dose male rats appears to be class-effect of carbonic anhydrase inhibitors in rats. Rats are particularly prone to developing papillomas in response to foreign bodies, compounds causing crystalluria and diverse sodium salts.

No changes in bladder urothelium were seen in dogs given oral dorzolamide hydrochloride for 1 year at doses of 2mg/kg/day or in monkeys given 20µL of 3% dorzolamide hydrochloride topically to the eye bid of 1 year.

Dorzolamide showed no mutagenic potential in a series of standard assays for gene mutations, chromosomal damage and DNA damage.

In reproduction studies of dorzolamide hydrochloride in rats, there were no adverse effects on the reproductive capacity of males or females at oral doses up to 15 and 7.5mg/kg/day.

Timolol Maleate

In a two-year oral study of timolol maleate in rats there was a statistically significant (p<0.05) increase in the incidence of adrenal pheochromocytomas in male rats administered 300 mg/kg/day (300 times** the maximum recommended human oral dose). Similar differences were not observed in rats administered oral doses equivalent to 25 or 100 times the maximum recommended human oral dose.

In a lifetime oral study in mice, there were statistically significant (p<0.05) increases in the incidence of benign and malignant pulmonary tumors, benign uterine polyps and mammary adenocarcinoma in female mice at 500 mg/kg/day (500 times the maximum recommended human dose), but not at 5 or 50mg/kg/day. In a subsequent study in female mice, in which post-mortem examinations were limited to uterus and lungs, a statistically significant increase in the incidence of pulmonary tumors was again observed at 500 mg/kg/day.

The increased occurrence of mammary adenocarcinoma was associated with elevations in serum prolactin which occurred in female mice administered timolol at 500 mg/kg, but not at doses of 5 or 50 mg/kg/day. An increased incidence of mammary adenocarcinomas in rodents has been associated with administration of several other therapeutic agents which elevate serum prolactin, but no correlation between serum prolactin levels and mammary tumors has been established in man. Furthermore, in adult human female subjects who received oral dosages of up to 60 mg of timolol maleate, the maximum recommended human oral dosage, there were no clinically meaningful changes in serum prolactin.

Timolol maleate was devoid of mutagenic potential when evaluated *in vivo* (mouse) in the micronucleus test and cytogenetic assay (doses up to 800 mg/kg) and *in vitro* in a neoplastic cell transformation assay (up to 100 mcg/ml). In Ames tests the highest concentrations of timolol employed, 5000 or 10,000 mcg/plate, were associated with statistically significant elevations (p<0.05) of revertants observed with tester strain TA 100 (in seven replicate assays), but not in the remaining three strains. In the assays with tester strain TA 100, no consistent dose-response relationship was observed, nor did the ratio of test to control revertants reach 2. A ratio of 2 is usually considered the criterion for a positive Ames test.

Reproduction and fertility studies in rats showed no adverse effect on male or female fertility at doses up to 150 times the maximum recommended human oral dose.

VII. PREGNANCY

There are no adequate and well-controlled studies in pregnant women. COSOPT should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

VIII. NURSING MOTHERS

It is not known whether dorzolamide hydrochloride is excreted in human milk. Timolol maleate does appear in human milk. Because of the potential for serious adverse reactions on the nursing infant, a decision should be made whether to discontinue nursing or discontinue the drug, taking into account the importance of the drug to the mother.

** The maximum recommended oral dose is 60 mg of timolol. One drop of TIMOPTOL 0.5% contains about 1/300 of this dose which is about 0.2 mg.

IX. PEDIATRIC USE

The safety and efficacy of 2% dorzolamide hydrochloride ophthalmic solution has been established in a clinical study of children under the age of 6 years. In this study patients under 6 and greater than 2 years of age whose IOP was not controlled with monotherapy received COSOPT. In those patients COSOPT was generally well tolerated.

X. USE IN ELDERLY

Of the total number of patients in clinical studies of COSOPT, 49% were 65 years of age and over, while 13% were 75 years of age and over. No overall differences in effectiveness or safety were observed between these patients and younger patients, but greater sensitivity of some older individuals cannot be ruled out.

XI. DRUG INTERACTIONS

Specific drug interaction studies have not been performed with COSOPT.

Page: 1 of 3

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GLAMS Part Number:

CAW-5799-02

Technical Drawing No:

N/A

Dimension:

387 X 240 mm

Pharma Code:

7852

2D matrix:

N/A

Date:

18-05-2023

Market:

MY

Material Code:

754751

Artwork Version Number:

03

Schawk Job No.:

108718471/403302982

Printable Colours:

Santen PI Blue

Non Print:

Profile

Technical Info

SCK

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In clinical studies, COSOPT was used concomitantly with the following systemic medications without evidence of adverse interactions: ACE-inhibitors, calcium channel blockers, diuretics, non-steroidal anti-inflammatory drugs including aspirin, and hormones (e.g., estrogen, insulin, thyroxine). However, the potential exists for additive effects and production of hypotension and/or marked bradycardia when timolol maleate ophthalmic solution is administered together with oral calcium channel blockers, catecholamine-depleting drugs, antiarrhythmics, parasympathomimetics or beta-adrenergic blocking agents. Potentiated systemic beta-blockade (e.g., decreased heart rate, depression) has been reported during combined treatment with CYP2D6 inhibitors (e.g. quinidine, SSRIs) and timolol.

The dorzolamide component of COSOPT is a carbonic anhydrase inhibitor and although administered topically, is absorbed systemically. In clinical studies, dorzolamide hydrochloride ophthalmic solution was not associated with acid-base disturbances. However, these disturbances have been reported with oral carbonic anhydrase inhibitors and have, in some instances, resulted in drug interactions (e.g., toxicity associated with high-dose salicylate therapy). Therefore, the potential for such drug interactions should be considered in patients receiving COSOPT.

Oral β-adrenergic blocking agents may exacerbate the rebound hypertension which can follow the withdrawal of clonidine.

Frequency of ocular inflammatory adverse reactions including conjunctival hyperemia was elevated in a clinical trial with concomitant administration of omdenepag isopropyl and timolol maleate. Therefore, concomitant use of omdenepag isopropyl with timolol maleate should be done with caution.

XII. SIDE EFFECTS

In clinical studies, COSOPT was generally well tolerated; no adverse experiences peculiar to this combination drug have been observed. Adverse experiences have been limited to those that were reported previously with dorzolamide hydrochloride and/or timolol maleate. In general, common adverse experiences were mild and did not cause discontinuation.

During clinical studies, 1035 patients were treated with COSOPT. Approximately 2.4% of all patients discontinued therapy with COSOPT because of local ocular adverse reactions. Approximately 1.2% of all patients discontinued because of local adverse reactions suggestive of allergy or hypersensitivity. (such as lid inflammation and conjunctivitis)

Like other topically applied ophthalmic medicines, timolol is absorbed into the systemic circulation. This may cause similar undesirable effects as seen with systemic beta-blocking agents. Incidence of systemic ADRs after topical ophthalmic administration is lower than for systemic administration.

The following adverse reactions have been reported with COSOPT or one of its component either during clinical trials or during post-marketing experience:

System Organ Class (MedDRA)	Formulation	Very Common	Common	Uncommon	Rare	Not Known**
Immune system disorders	<COSOPT>				signs and symptoms of systemic allergic reactions, including angioedema, urticaria, pruritus, rash, anaphylaxis	
	Timolol maleate eye drops, solution				signs and symptoms of allergic reactions including angioedema , urticaria, localized and generalized rash, anaphylaxis	pruritus
Metabolism and nutrition disorders	Timolol maleate eye drops, solution					hypoglycaemia
Psychiatric disorders	Timolol maleate eye drops, solution			depression*	insomnia*, nightmares*, memory loss	hallucination
Nervous system disorders	Dorzolamide hydrochloride eye drops, solution		headache*		dizziness*, paraesthesia*	
	Timolol maleate eye drops, solution		headache*	dizziness*, syncope*	paraesthesia*, increase in signs and symptoms of myasthenia gravis, decreased libido*, cerebrovascular accident*, cerebral ischaemia	
Eye disorders	<COSOPT>	burning and stinging	conjunctival injection, blurred vision, corneal erosion, ocular itching, tearing			
	Dorzolamide hydrochloride eye drops, solution		eyelid inflammation*, eyelid irritation*	iridocyclitis*	irritation including redness*, pain*, eyelid crusting*, transient myopia (which resolved upon discontinuation of therapy), corneal oedema*, ocular hypotony*, choroidal detachment (following filtration surgery)*	foreign body sensation in eye
	Timolol maleate eye drops, solution		signs and symptoms of ocular irritation including blepharitis*, keratitis*, decreased corneal sensitivity, and dry eyes*	visual disturbances including refractive changes (due to withdrawal of miotic therapy in some cases)*	ptosis, diplopia, choroidal detachment (following filtration surgery)* (see Special warning and precautions for use 4.4)	itching, tearing, redness, blurred vision, corneal erosion
Ear and labyrinth disorders	Timolol maleate eye drops, solution				tinnitus*	

System Organ Class (MedDRA)	Formulation	Very Common	Common	Uncommon	Rare	Not Known**
Cardiac disorders	Timolol maleate eye drops, solution			bradycardia*	chest pain*, palpitation*, oedema*, arrhythmia*, congestive heart failure*, cardiac arrest*, heart block	atrioventricular block, cardiac failure
	Dorzolamide hydrochloride eye drops, solution					palpitations
Vascular disorders	Timolol maleate eye drops, solution				hypotension*, claudication, Raynaud's phenomenon*, cold hands and feet*	
Respiratory, thoracic, and mediastinal disorders	<COSOPT>		sinusitis		shortness of breath, respiratory failure, rhinitis, rarely bronchospasm	
	Dorzolamide hydrochloride eye drops, solution				epistaxis*	dyspnoea
	Timolol maleate eye drops, solution			dyspnoea*	bronchospasm (predominantly in patients with pre-existing bronchospastic disease)*, respiratory failure, cough*	
Gastrointestinal disorders	<COSOPT>	dysgeusia				
	Dorzolamide hydrochloride eye drops, solution		nausea*		throat irritation, dry mouth*	
	Timolol maleate eye drops, solution			nausea*, dyspepsia*	diarrhea, dry mouth*	dysgeusia, abdominal pain, vomiting
Skin and subcutaneous tissue disorders	<COSOPT>				contact dermatitis, Stevens-Johnson syndrome, toxic epidermal necrolysis	
	Dorzolamide hydrochloride eye drops, solution				rash*	
	Timolol maleate eye drops, solution				alopecia*, psoriasiform rash or exacerbation of psoriasis*	skin rash
Musculoskeletal and connective tissue disorders	Timolol maleate eye drops, solution				systemic lupus erythematosus	myalgia
Renal and urinary disorders	<COSOPT>			urolithiasis		
Reproductive system and breast disorders	Timolol maleate eye drops, solution				Peyronie's disease*, decreased libido	sexual dysfunction
General disorders and administration site conditions	Dorzolamide hydrochloride eye drops, solution		asthenia/ fatigue*			
	Timolol maleate eye drops, solution			asthenia/ fatigue*		

* These adverse reactions were also observed with <COSOPT> during post-marketing experiences.

** Additional adverse reactions have been seen with ophthalmic beta-blockers and may potentially occur with <COSOPT>.

XIII. OVERDOSAGE

No data are available with regard to human overdosage by accidental or deliberate ingestion of COSOPT.

There have been reports of inadvertent overdosage with timolol maleate ophthalmic solution resulting in systemic effects similar to those seen with systemic beta-adrenergic blocking agents such as dizziness, headache, shortness of breath, bradycardia, bronchospasm, and cardiac arrest. The most common signs and symptoms to be expected with overdosage of dorzolamide are electrolyte imbalance, development of an acidotic state, and possibly central nervous system effects.

Treatment should be symptomatic and supportive. Serum electrolyte levels (particularly potassium) and blood pH levels should be monitored. Studies have shown that timolol does not dialyze readily.

XIV. AVAILABILITY

COSOPT is available in single packs of 5mL.

COSOPT is a clear, colorless to nearly colorless, slightly viscous solution.

XV. STORAGE

Store below 30°C. Protect from light. Discard one month after first opening.

XVI. SHELF LIFE

Please refer to the expiry date on the outer carton.

XVII. MANUFACTURER & PACKER

SANTEN PHARMACEUTICAL CO., LTD.

Noto plant: Ishikawa, Japan

XVIII. PRODUCT REGISTRATION HOLDER

SANTEN PHARMA MALAYSIA SDN. BHD.

(1117719-T)

Unit #23A-10, Q Sentral, No. 2A, Jalan Stesen Sentral 2, Kuala Lumpur Sentral, 50470 Kuala Lumpur, Malaysia.

DATE OF REVISION: September 2021

Username	Full Name	Status	Date/Time (UTC)
GaithriGanesan	Gaithri Ganesan	Approved	18-May-2023 11:30 UTC
CalvinLow	Calvin Low	Approved	18-May-2023 11:59 UTC
KahPeiLau	Kah Pei Lau	Approved	18-May-2023 13:16 UTC
GraceLou	Grace Lou	Approved	19-May-2023 06:03 UTC
VonnyHuang	Vonny Huang	Approved	19-May-2023 10:20 UTC
KazuhiroNishino	Kazuhiro Nishino	Approved	19-May-2023 10:25 UTC